

UDDISH VIRALKUMAR PATEL

Software Engineer & Cybersecurity Specialist

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PROFESSIONAL SUMMARY

Software Engineer & Cybersecurity Specialist with expertise in full-stack development, distributed systems, and application security. Hands-on experience in vulnerability assessment, web security testing, REST API design, database optimization, and production deployment. Proficient in building secure, scalable applications from architecture to deployment. Strong foundation in OWASP security practices, network fundamentals, and systems engineering.

TECHNICAL SKILLS

Security & Web Application: Vulnerability Assessment | Penetration Testing | OWASP Top 10 | XSS/SQLi Detection | Security Headers Analysis | Input Validation | IDS/WAF Concepts | HTTP Traffic Analysis | Risk Assessment

Programming & Development: Python | JavaScript | HTML5 | CSS3 | REST API Design | Performance Optimization | Error Handling

Backend & Frameworks: Flask | SQLAlchemy | .NET Basics | API Architecture | Routing & Middleware

Databases & Systems: PostgreSQL | SQLite | Database Optimization | Query Tuning | Schema Design | Indexing | Database Migration

Cloud & DevOps: Docker | Railway | Git/GitHub | CI/CD | Streamlit | Environment Configuration | Container Orchestration

Network & Infrastructure: TCP/IP | Port Scanning | DNS/WHOIS | HTTP/HTTPS Lifecycle | Network Communication | Subdomain Enumeration

Tools & Platforms: TryHackMe | Git & GitHub | Linux CLI | Vulnerability Scanners | Visual Studio Code | Postman

TECHNICAL PROJECTS

1. Password Cracking & Credential Attack Suite - Security Toolkit

Developed an ethical password security toolkit with live interactive dashboard for real-world hash cracking simulations. Implemented crypto-hash cracking against Linux shadow (\$1/\$5/\$6\$) and Windows NTLM hashes. Built live hardware benchmark measuring actual hashing performance (hashes/sec) at runtime. Integrated NIST SP 800-63B compliance engine with breach corpus analysis, entropy calculation, and pattern-based vulnerability checks. Created visual analytics: entropy histogram, severity pie chart, risk ranking. Implemented 38 automated pytest tests. Deployed on Streamlit Community Cloud.

2. SentinelShield - Intrusion Detection System & Web Application Firewall

Engineered educational IDS/WAF system analyzing HTTP traffic and detecting web attacks (SQL injection, XSS). Implemented packet inspection, real-time alert generation, and dashboard visualization. Built secure request logging and efficient query performance using Flask + PostgreSQL. Designed data pipelines for security event logging and analysis. Demonstrated network protocol understanding and traffic monitoring at scale.

3. AegisScan - Web Vulnerability Scanner

Designed modular distributed architecture with separate backend modules: DNS resolution, port scanning, API discovery, directory enumeration. Implemented scalable REST API handling concurrent scan requests with async result processing. Built modules detecting XSS and SQL Injection vulnerabilities. Created interactive dashboard for result visualization. Deployed with Docker containerization for service isolation and scalable deployment patterns.

4. CA Management System - Full-Stack Production Application

Architected and deployed production-grade client management system using Flask + SQLAlchemy. Optimized database: migrated SQLite -> PostgreSQL for production reliability. Designed RESTful APIs with robust error

handling and data consistency. Deployed containerized application using Docker + Railway with environment configuration and secrets management. Demonstrated systems-level thinking for load handling and concurrent requests.

5. Doctor Appointment System - Full-Stack Web Application

Built end-to-end web application for appointment scheduling with role-based access control. Frontend: Responsive HTML5/CSS3/JavaScript UI. Backend: Flask API with proper state management and authentication focus. Designed database for efficient scheduling queries and appointment consistency. Evaluated authentication mechanisms and implemented validation techniques to reduce security risks.

EDUCATION

B.E. Computer Science & Engineering | CGPA: 7.74/10

Narnarayan Shastri Institute of Technology (NSIT), Gujarat Technological University (GTU)

TRAINING & CERTIFICATIONS

SkillLogic Cybersecurity Course - Ethical hacking, vulnerability concepts, security tools and methodologies

TryHackMe - 20% Completed: Hands-on labs (reconnaissance, web attacks), real-world attack simulations

KEY STRENGTHS

Security & Systems Mindset: End-to-end application design from architecture to deployment with security-first thinking

Production-Ready Code: Built and deployed real applications; familiar with deployment, monitoring, and debugging challenges

Full-Stack Capabilities: Comfortable across frontend, backend, databases, DevOps, and security testing

Fast Learner: Self-taught Docker, PostgreSQL optimization, cloud deployment, and advanced security tools

Independent Problem Solver: Autonomously own projects from ideation through production; strong debugging and troubleshooting skills